Bartlett adjustment for random effects meta-analysis of event outcomes with small samples

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Abstract. ... Keywords: Meta-analysis, Bartlett correction, Generalized linear mixed models

1. introduction: (skip generic metaanalysis material). since stijnen 2010, glmm have become the preferred model for random effects meta-analysis based on count data. cochran etc have recommended ... . cite elsewhere for advantages over inverse variance model. while the glmm approach avoids many of the dubious assumptions of the inverse-variance approach, inference is still based on a chi squared asymptotic expansion. therefore as in the IV case the conclusions based on smaller meta-analyses are suspect. bartlett correction leverages modelling assumptions to improve the efficiency of the testing procedure. when the modelling assumptions are met, the imrpovement is typically a jump in the rate of convergence from 1/n to 1/n2. [[the key modelling assumption in this setting is the gaussianity of the random effect.]]

while bartlett corrections were long ago applied to glm's, there do not appear to be extensions to even the simple glmm's that are used in meta-analysis. the reason for the reluctance appears to be computational difficulty [[cite hanada]]. unlike glms closed form expressions are unavailable for glmm corrections. we argue that with careful design these difficulties can be overcome and consumer-grade hardware is sufficient for practical implementation.

paper outline. give stifjnen's glmm approach as background, then describe algo. next look at sims, focusing on robustness. for data analysis walk through the 4 stijnen analyses.

2. background.

2a. notation for events, trials. following stijnen we consider 4 types of analyses: single arm and two arm trials, proportions and incidence/rates. let n denote the number of studies included in the meta-analysis. for single-arm trials let yi denote the number of events in study i, and for two-arm trials let yti and yci denote the number of events in the treatment and control arms of study i. when propostions are involved let mi denote the total number of events in study i, or mti and mci in case of a two-arm trial. when incidence rates are involved let Ti, or Tti and Tci, denote the period of time during which events are counted. see ((data anlaysis section)) for concrete examples.[[maybe better if notation can b eintroduced in the course of the 4 examples...]]

in all casea a random effect models variation between the effects of the studies. this

random effect is modelled as gaussian:

$$\theta_i = \mu_0 + \tau_0 Z, Z \sim N(0,1)$$

the target of inference in the meta-analysis is μ_0, with τ_0^2 a nuisance parameter.

2a-1 single arm, proportion: binomial-prop. each study reports the number of events yi and number of units mi. the target of inference μ_0 represents the log-odds of an event. the statistical model is binomial

$$f(Y_i \mid Z) = ...$$

mi is treated as non-random.

2a-2 two arm, proportion: each study reports the number of treatment group events yti and control group events yci and the total number of units mti and mci. the target of inference is the ratio of the treatment and control log-odds. the statistical model conditions on the total number of events si=yti+yci, under which yti follows fisher's non-central hypergeometric distribution

$$f(yti \mid theta_i, s_i) = ...$$

2a-3 single arm, incidence: poisson. each study reports the number of events and the duration of observation Ti. the target of inference represents the log of the incidence rate. the model is poisson

$$f(yti \mid theta_i) =$$

the follow-up time Ti is treated as non-random.

2a-4 two arm, incidence rate ratio: binomial-irr. each study reports the number of treatment group events yti and control group events yci and the durations of observation Tti and Tci. the target of inference is the log of the ratio of the incidence rates between the two arms. the model is binomial

$$f(yti \mid theta_i) =$$

As before the periods Tti and Tci are teated as non-random.

in all cases the likelihood for study i has the form:

$$L(\theta, \tau^2 | Y_i) = \int_{-\infty}^{\infty} f(y|z)\phi(z)dz$$

where $f(y|z)$ is the conditional PMF given the random effect, as given above.

The LR test is based on the test statistic

$$W = \sum_{i=1}^{n} ...$$

where ... . [[copy from other ms]] By Wilks theorem W typically converges to a chi squared distribution with a single degree of freedom when the null hypothesis holds. Relying on this

asymptotic approximation a confidence interval for μ_0 can be formed by inverting the test statistic.[[is that what the profile method is doing??]]

2b. bartlett correction

bartlett correction. under a range of conditions, chiefly the existence of an edgeworth expansion for the underlying random variables, the mean LR statistic admits an expansion in integral powers of $1/n$

$$\mathbb{E}(W) =$$

It follows that

$$defnbartlettcorrectedstatistic$$

has a bias of order $O(1/n^2)$. the work of ((lawley and bartlett)) shows that not only does ((ref above)) match the mean of a chi squared to order $O(1/n^2)$, but also all other moments, and so the entire distribution. when the above assumptions ((mention assumptions for wilks, the mean expansion, and lawley)) hold the resulting asymptotic rate of convergence is 1/n2. may be used with profile curve to obtain CIs and p-vals. ((cite to sim/data analysis.))

two main voilations of the assumptions t oconsider: gaussian random effect and discreteness of the data. important exception to the "normal course" is lattice data such as considered here. expansion ((lr expansion)) may not be justified. literature unclear. bartlett corrections for discrete glm's have been developed.

3. proposed method.

lawley ((cite)) provides expressions for the correction ((ref)) as sums and products of the components of the inverse fisher information and certain tensors [[k terms]], defined as follows. letting ... [[define inner only k]]. the information matrix is then ... . terms of the form ..., are required for the computation of the correction. taking into account permutation symmetry of the order of differentiation, there are 35 k terms in a model with 1 parameter of interest and 1 nuisance parameter, as considered here.

computing the bartlett correction ((ref)) comes down to computing the tensor components ((ref)) for the model ((ref)). we begin by approximating the integral by a quadrature rule. since all the likelihoods involve integrating against a gaussian, gauss-hermite approximation is a natural choice, replacing the integral ((ref)) with a weighted sum the weights ... . ((how much detail to give here, maybe just leave it as a sum approximation.))

$$L(\theta, \tau^2|Y) = \int_{-\infty}^{\infty} f(Y|z)\phi(z)dz \approx \sum_j w_j f(Y|z_j)$$

let r denot eth emultivector, the derivative of the log likelihood for a unit wrt to r is

$$D_r \log L(\theta, \tau^2|Y) \approx D_r \log \sum_j w_j f(Y|z_j)$$

Its expectation may be approximated by another quadrature

$$\begin{aligned} k_r &\approx \sum_y \left(D_r \log \sum_j w_j f(y|z_j) \right) \int f(y|z)\phi(z)dz \\ &\approx \sum_y \left(D_r \log \sum_j w_j f(y|z_j) \right) \sum_k w_k f(y|z_k) \end{aligned}$$

The outer sum is taken over the support of Y, which is a finite set in all the models considered here other than the Poisson model. The Poisson model requires a truncation rule, discussed further below. the general tensor component is then

$$k_r^{(s)} \approx \sum_y D_s \left(\left(D_r \log \sum_j w_j f(y|\theta_j) \right) \sum_k w_k f(y|\theta_k) \right)$$

where as before s is a multivector of indices.

carrying these differentiations through yields rational expressions in derivatives of the weighted sums $\sum_j w_j f(y|z_j)$. The form of these 35 expressions, for the 35 k terms, are the same for any model based on a likelihood of the form ((ref)), and may be pre-computed. for example ... and ....

The derivatives of the model-specific conditoinal pmf's that appear in the 35 expressions number 15: ... . the form of these derivatives may be pre-computed but their specific values in general depend on parameters of the conditional pmf ((see above)), and must usually be instantiated for each study in the meta-analysis. these derivatives may be re-written using well-known recurrences to polynomial expressions of the conditional pmf itself, for which fast and numerically stable routines are available. for example: ... . [[see appendix for similar recurrences, or refer to johnson and kotz or wherever]]

further implementation details.

chain rule: differentiate wrt pmf sums. get 35 k terms in terms of pmfsums (give an example or two).

differentiate pmfs using recurrences in terms usual pmfs to allow fast, stable routines to be used.

some specific issues: support for poisson. speed issues with hyperg.

3. simulation

goals: robustness/gaussianity check. discreteness issue.

4. data analysis.

4a. proportions

4b. incidence